Functions

Worksheet 2 (Calculator)

1. W17/qp42/q9

$$f(x) = 1 - 2x \qquad g(x) = x + 4 \qquad h(x) = x^2 + 1$$

a. Find $f(-1)$. [1]

b. Solve the equation $2f(x) = g(x)$ [2]

c. Find $fg(x)$. Give your answer in its simplest form. [2]

d. Find $hh(2)$. [2]

e. Find $f^{-1}(x)$. [2]

f. $hfg(x) = 4x^2 + px + q$

Find the value of p and the value of q. [4]

2. S18/qp42/q8

$$f(x) = 8 - 3x \qquad g(x) = \frac{10}{x+1} \quad x \neq -1 \qquad h(x) = 2^x$$

a. Find

i. $hf(\frac{8}{3})$ [2]

ii. $gh(-2)$ [2]

iii. $g^{-1}(x)$ [3]

iv. $f^{-1}f(5)$ [1]

b. Write $f(x) + g(x)$ as a single fraction in its simplest form. [3]

3. S18/qp43/q10

a. $f(x) = 8 - 3x$ $\qquad g(x) = \frac{10}{x+1} \quad x \neq -1$

i. Find $gg(2)$. [2]

ii. Find $g(x+2)$, giving your answer in its simplest form. [2]

iii. Find x when $f(x) = 7$. [2]

iv. Find $f^{-1}(x)$. [2]

b. $h(x) = x^x,\ x > 0$

i. Calculate $h(0.3)$. Give your answer correct to 2 decimal places. [2]

ii. Find x when $h(x) = 256$. [1]

4. W18/qp43/q9

$f(x) = 3x + 4 \qquad g(x) = 2x - 1 \qquad h(x) = 3^x$

a. Work out $g\left(\frac{1}{2}\right)$. [1]

b. Work out $fh(-1)$. [2]

c. Find $g^{-1}(x)$. [2]

d. Find $ff(x)$, giving your answer in its simplest form. [2]

e. Find $\left(f(x)\right)^2$ in the form $ax^2 + bx + c$. [2]

f. Find x when $h^{-1}(x) = g(2)$. [2]

5. M19/qp42/q8

$$f(x) = \frac{3}{x+2} \quad x \neq -2 \qquad g(x) = 8x - 5 \qquad h(x) = x^2 + 6$$

a. Work out $g\left(\frac{1}{4}\right)$. [1]

b. Work out $ff(2)$. [2]

c. Find $gg(x)$, giving your answer in its simplest form. [2]

d. Find $g^{-1}(x)$. [2]

e. Write $g(x) - f(x)$ as a single fraction in its simplest form. [3]

f.

i. Show that $hg(x) = 19$ simplifies to $16x^2 - 20x + 3 = 0$. [3]

ii. Use the quadratic formula to solve $16x^2 - 20x + 3 = 0$. Show all your working and give your answers correct to 2 decimal places. [4]

Answer Key

1. W17/qp42/q9

9(a)	3	**1**	
9(b)	$-\frac{2}{5}$ oe	**2**	**M1** for $2(1-2x)=x+4$
9(c)	$-2x-7$ final answer	**2**	**M1** for $1 - 2(x + 4)$
9(d)	26	**2**	**B1** for h(5) soi or **M1** for $\left(x^2+1\right)^2+1$
9(e)	$\frac{1-x}{2}$ oe final answer	**2**	**M1** for $x = 1 - 2y$ or $2x = 1 - y$ or $\frac{y}{2}=\frac{1}{2}-x$ or $y - 1 = -2x$
9(f)	$[p =] -20$ $[q =]$ 26	**4**	**B3** for [hgf(x)] = $4x^2-20x+26$ seen and not spoilt by further working or **M1** for $(1 - 2x) + 4$ **M1 dep** for $(\textit{their}\ (5-2x))^2+1$ **B1FT dep** for $25 - 10x - 10x + 4x^2$

2. S18/qp42/q8

8(a)(i)	1	**2**	**M1** for h(0) or for 2^{8-3x}
8(a)(ii)	8	**2**	**M1** for g(¼) or for $\frac{10}{2^x+1}$
8(a)(iii)	$\frac{10-x}{x}$ or $\frac{10}{x}-1$ final answer	**3**	**M2** for $x = \frac{10-y}{y}$ or better or $xy = 10 - x$ or better or $y + 1 = \frac{10}{x}$ or **M1** for $x(y + 1) = 10$ or $y(x + 1) = 10$ or $x = \frac{10}{y+1}$ or $x + 1 = \frac{10}{y}$
8(a)(iv)	5	**1**	
8(b)	$\frac{-3x^2+5x+18}{x+1}$ final answer	**3**	**M1** for $\frac{(8-3x)(x+1)+10}{x+1}$ **B1** for $-3x^2 - 3x + 8x + 8$ [+10]

3. S18/qp43/q10

10(a)(i)	26	**2**	**M1** for g(5) or for $\left(x^2+1\right)^2+1$
10(a)(ii)	x^2+4x+5	**2**	**M1** for $\left(x+2\right)^2+1$
10(a)(iii)	5	**2**	**M1** for $2x-3=7$
10(a)(iv)	$\frac{x+3}{2}$ oe	**2**	**M1** for $x=2y-3$ or $y+3=2x$ or $\frac{y}{2}=x-\frac{3}{2}$ oe
10(b)(i)	[0].70 cao	**2**	**B1** for [0].696 to [0].697
10(b)(ii)	4 cao	**1**	

4. W18/qp43/q9

9(a)	0	**1**	
9(b)	5	**2**	**M1** for $3(3^x)+4$ or better or $f(\frac{1}{3})$ or $f(3^{-1})$
9(c)	$\frac{x+1}{2}$ oe final answer	**2**	**M1** for $x=2y-1$ or $y+1=2x$ or $\frac{y}{2}=x-\frac{1}{2}$ or better
9(d)	$9x+16$	**2**	**M1** for $3(3x+4)+4$ oe
9(e)	$9x^2+24x+16$	**2**	**B1** for three terms from $9x^2+12x+12x+16$ correct
9(f)	27	**2**	**M1** for x = h(*their* g(2))

5. M19/qp42/q8

8(a)	-3	**1**	
8(b)	$\frac{12}{11}$ oe	**2**	**M1** for $\frac{3}{\frac{3}{x+2}+2}$ soi
8(c)	$64x-45$ final answer	**2**	**M1** for $8(8x-5)-5$ isw
8(d)	$\frac{x+5}{8}$ oe final answer	**2**	**M1** for a correct first step $y+5=8x$, $\frac{y}{8}=x-\frac{5}{8}$ or $x=8y-5$
8(e)	$\frac{8x^2+11x-13}{x+2}$ final answer	**3**	**M1** for $(8x-5)(x+2)-3$ oe isw **B1** for common denominator $(x+2)$
8(f)(i)	$(8x-5)^2+6 = 19$	**M1**	
	$64x^2-40x-40x+25$	**B1**	
	$64x^2-40x-40x+25+6=19$ oe leading to $16x^2-20x+3=0$	**A1**	with no errors and must show $(8x-5)^2+6 = 19$ with no omissions after this
8(f)(ii)	$\frac{[--]20\pm\sqrt{([-]20)^2-4(16)(3)}}{2\times16}$ oe	**2**	**B1** for $\sqrt{([-]20)^2-4(16)(3)}$ or better or **B1** for $\frac{[--]20+\sqrt{q}}{2(16)}$ oe or $\frac{[--]20-\sqrt{q}}{2(16)}$
	0.17 and 1.08 final ans	**2**	**B1** for each If 0 scored, **SC1** for answer 0.2 and 1.1 or answer – 0.17 and –1.08 or 0.174... and 1.075 to 1.076 seen or 0.17 and 1.08 seen in working